

DaSEH Day 1 Cheatsheet

Basic R

Major concepts

- **Package:** A bundle of code and/or data that can be loaded for repeated use or sharing, like an expansion pack.
- **Object:** Something that can be worked with in R.
- **Function:** A piece of code that performs a task in R. Functions can come from base R, additional packages, or your own code.
- **Argument:** An option specified in a function.
- **tidyverse:** A set of packages designed for data science that can make code more intuitive than base R.
- **R console:** A full calculator:
 - +, -, /, and * add, subtract, divide, and multiply.
 - ^ or ** raises to a power.
 - Parentheses, (and), control order of operations.
 - %% finds the remainder.
- **Comments:** # is the comment symbol; nothing to its right is evaluated.

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	<code><-</code>	<code>x <- 1</code>	Assigns a name, <code>x</code> , to something in the R environment.
Base R	<code>c()</code>	<code>x <- c(1, 3)</code>	Combines values into a vector.
Base R	<code>str()</code>	<code>str(x)</code>	Summarizes the structure of object <code>x</code> .
Base R	<code>class()</code>	<code>class(x)</code>	Returns the type of values in object <code>x</code> .
Base R	<code>length()</code>	<code>length(x)</code>	Returns the length of object <code>x</code> .
Base R	<code>seq()</code>	<code>seq(from = 0, to = 100, by = 5)</code>	Generates regular sequences.
Base R	<code>rep()</code>	<code>rep(1, times = 10)</code>	Replicates values. Can use the <code>times</code> or <code>length.out</code> argument.

Library/Package	Piece of code	Example of usage	What it does
Base R	<code>sample()</code>	<code>sample(1:12)</code>	Samples from values, with or without replacement. <code>replace = TRUE</code> samples with replacement.
Base R	<code>install.packages()</code>	<code>install.packages("tidyverse")</code>	Installs packages.
Base R	<code>library()</code>	<code>library(tidyverse)</code>	Loads and attaches packages to the R environment. Run each time you start R.

RStudio

Major concepts

- **RStudio:** An Integrated Development Environment (IDE) for R that makes R easier to use.
- **Source/Editor:** Where code is written and saved in an `.R` script.
- **R Console:** Where code is executed. Code entered here is not saved on disk.
- **Workspace/Environment:** Shows objects currently loaded in R.
- **R Markdown:** `.Rmd` files that generate reports with code and output.
- **R Project:** Helps organize work, manage working directories, and track the active project.
- **Getting help:** For any function, use `?FUNCTION_NAME` or `help("FUNCTION_NAME")` to view its help file.
- **RStudio keyboard shortcuts:** [RStudio keyboard shortcuts](#)

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	View()	<code>View(mtcars)</code>	Opens data in a spreadsheet-like viewer.
Base R	head()	<code>head(mtcars)</code>	Returns the first 6 rows by default. Use the <code>n</code> argument to specify the number of rows.
Base R	tail()	<code>tail(mtcars)</code>	Returns the last 6 rows by default. Use the <code>n</code> argument to specify the number of rows.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.